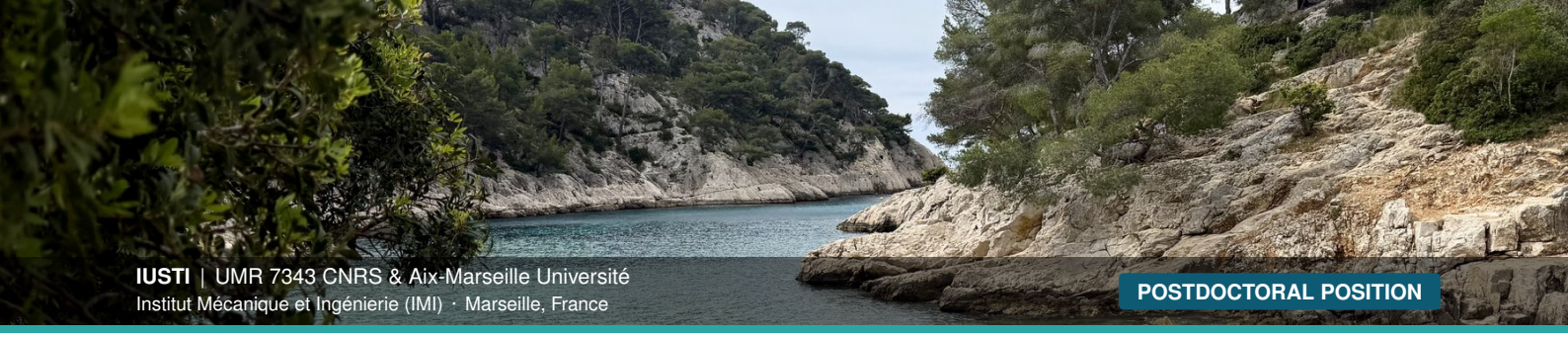


SNSF project *Viscocrack* · Aim 3

How do nonlinearly viscoelastic cracks select their paths?

Mixed-mode fracture in soft, dissipative solids

* * *

POSITION	Postdoctoral researcher, experimental	**DURATION**	18 months (project ends August 2028)	**START**	As soon as possible
PARTNERS	UCLA (L. Jin, phase-field modeling) · EPFL (E. Amstad, materials) · IRPHE (M.-J. Dalbe) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. When a crack is loaded by a combination of opening (Mode I) and in-plane shear (Mode II), it kinks and then follows a curved path. In brittle, linearly elastic solids, this path is predicted by local criteria such as the maximum hoop stress, the maximum energy release rate, or the principle of local symmetry [1,2]. Whether these criteria hold in materials whose response depends on loading rate and loading history is an open question: viscoelastic dissipation and large deformations near the tip reorganize the near-tip fields [3,4], so the same geometry may select different paths depending on how fast it is loaded. This postdoctoral position, dedicated to Aim 3 of *Viscocrack*, will measure crack path selection in quasi-two-dimensional (thin-sheet) specimens under controlled mixed-mode loading, varying the mode mixity and the loading rate relative to the material's relaxation times. Samples will be prepared in the lab from a well-established tough-hydrogel chemistry and from novel granular elastomers developed with the Soft Materials Laboratory at EPFL. Crack trajectories, kink angles, and near-tip strain fields will be resolved by high-speed imaging and digital image correlation, and the data will be used with the Jin group at UCLA to calibrate and test phase-field models of fracture in nonlinearly viscoelastic solids [5].

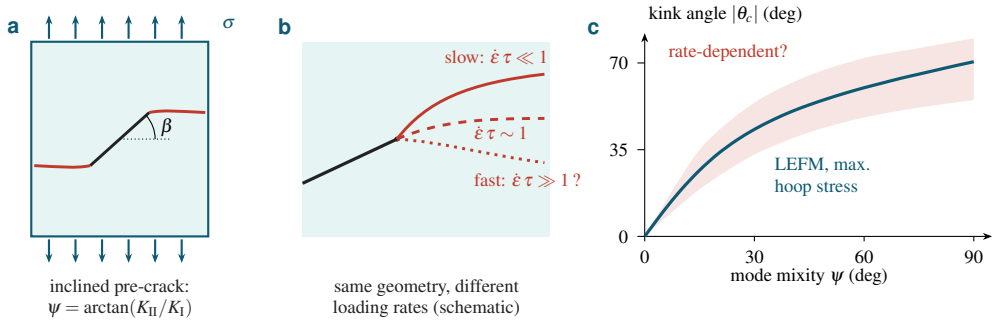


Figure 1. Mixed-mode path selection (schematic). (a) A pre-crack inclined to the load is loaded in combined Mode I and II and kinks out of its plane. (b) Hypothesis: in a nonlinearly viscoelastic solid, the selected path depends on the loading rate $\dot{\varepsilon}$ relative to a relaxation time τ. (c) The linear elastic kink angle (maximum hoop stress [1]) is the baseline against which rate-dependent selection will be measured.

Context. This position is part of the SNSF project *Viscocrack*, "Crack path prediction and control in nonlinearly viscoelastic materials: in-silico to experiments with viscoelastic and tough hydrogels." Tough hydrogels, elastomers, and soft tissues are increasingly built into medical devices and soft robots, yet we cannot reliably predict where a crack in such a material will go. *Viscocrack* addresses this question through four coordinated aims: materials development and characterization at EPFL (Soft Materials Laboratory, Prof. E. Amstad), phase-field modeling at UCLA (Prof. L. Jin), and experiments on mixed-mode path selection and on crack-path control at Aix-Marseille Université. The experimental work will be carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, with high-speed imaging set-ups at the Château-Gombert campus, in collaboration with Marie Julie Dalbe (IRPHE). IUSTI, IRPHE and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged. Exchanges and research visits with EPFL and UCLA are funded by the project.

Candidate profile. We seek a candidate with a PhD in physics, mechanics, materials science, or a related field and a strong experimental background, ideally in fracture, soft matter, or rheology. Experience with high-speed imaging, digital image correlation, or hydrogel and elastomer synthesis is an asset, as is fluency in quantitative image analysis (Python or MATLAB). The postdoc will lead the experimental program of Aim 3 and work day to day with a PhD student on the same aim, so enthusiasm for mentoring and for close exchange between experiments, theory, and simulation is expected. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] F. Erdogan and G. C. Sih, "On the crack extension in plates under plane loading and transverse shear," *J. Basic Eng.* **85**, 519–525 (1963).
[2] B. Cotterell and J. R. Rice, "Slightly curved or kinked cracks," *Int. J. Fract.* **16**, 155–169 (1980).
[3] C. Creton and M. Ciccotti, "Fracture and adhesion of soft materials: a review," *Rep. Prog. Phys.* **79**, 046601 (2016).
[4] O. Ronsin, C. Caroli, and T. Baumberger, "Crack front échelon instability in mixed mode fracture of a strongly nonlinear elastic solid," *EPL* **105**, 34001 (2014).
[5] B. Bourdin, G. A. Francfort, and J.-J. Marigo, "Numerical experiments in revisited brittle fracture," *J. Mech. Phys. Solids* **48**, 797–826 (2000).